# Laboratory Work No. 2

# Vector Control of PMSM

## Variant 3

Goal: to study field-oriented control of PMSM and tune PI speed and current regulators to satisfy the performance limits given in the assignment.

# 1. PMSM and FOC Model

The PMSM model from Lab 1 is controlled in the dq frame. The d-axis current reference is set to zero, while the q-axis current reference is produced by the outer PI speed controller. Decoupling terms are included to compensate cross-coupling and back-EMF.

| Parameter | Value |
|---|---|
| Variant | 3 |
| L, H | 0.033000 |
| R, Ohm | 5.3000 |
| Jm, kg*m^2 | 0.000600 |
| Pole pairs np | 5 |
| lambda_m, Wb | 0.1600 |
| B, N*m*s/rad | 0.0000 |

| Quantity | Value |
|---|---|
| Kp_i | 18.000000 |
| Ki_i | 1400.000000 |
| Kp_omega | 0.360000 |
| Ki_omega | 18.000000 |
| Current limit, A | 25.00 |
| Voltage limit, V | 120.00 |

# 2. Simulation Scheme

The control structure follows the classical cascade FOC design: speed reference -> PI speed controller -> current PI controllers in dq -> inverse Park transformation -> PMSM.

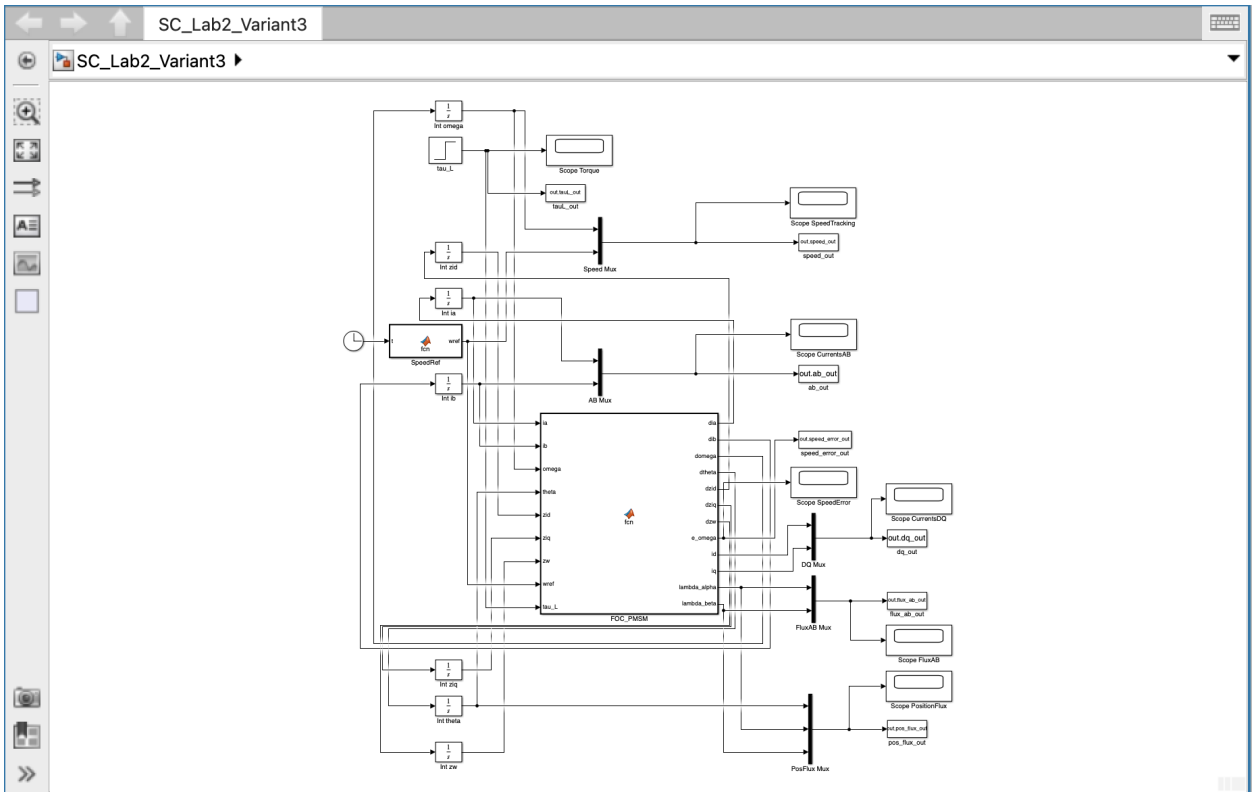


Figure 1. Main Simulink scheme of the PMSM vector-control system.

# 3. Simulink Screenshots and Transient Responses

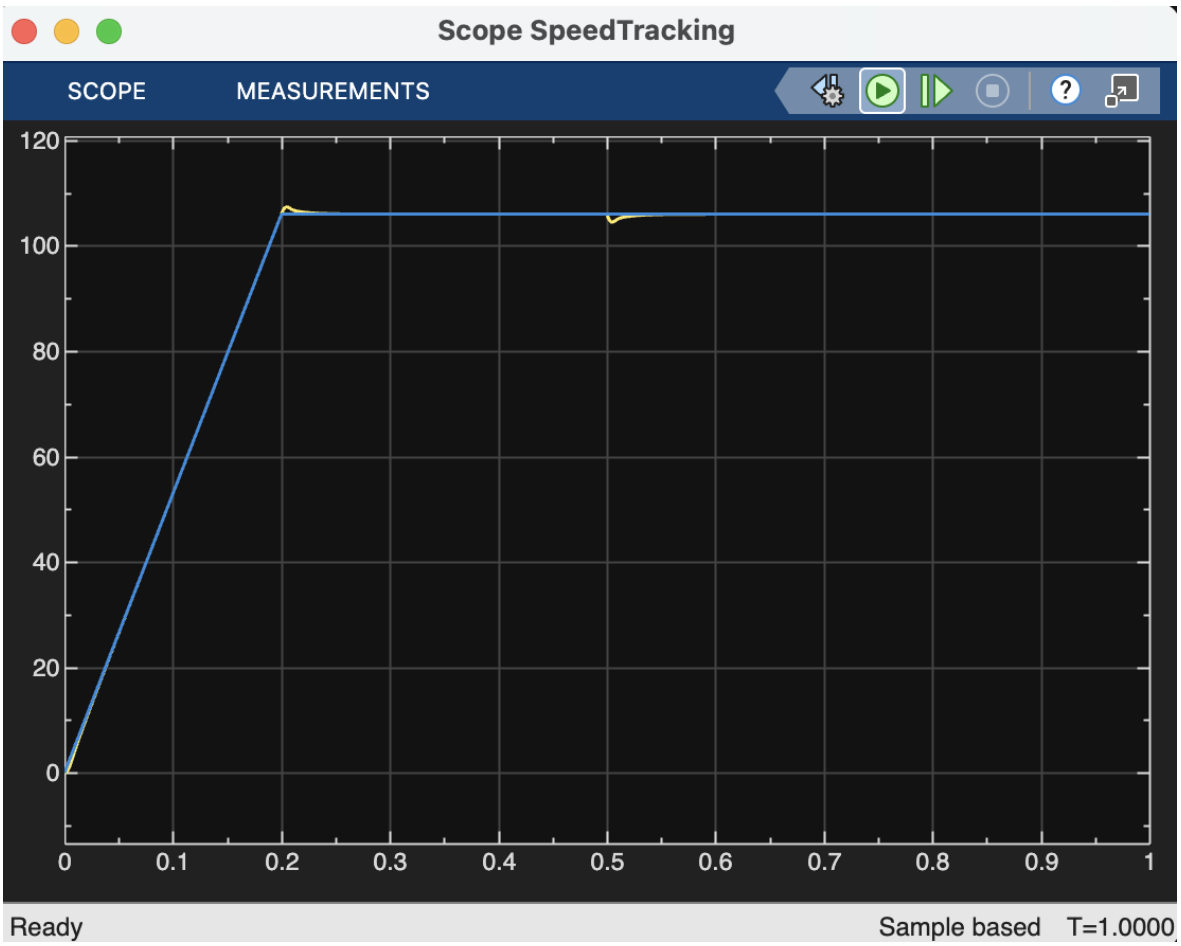


Figure 2. Scope of the actual speed and the reference speed.

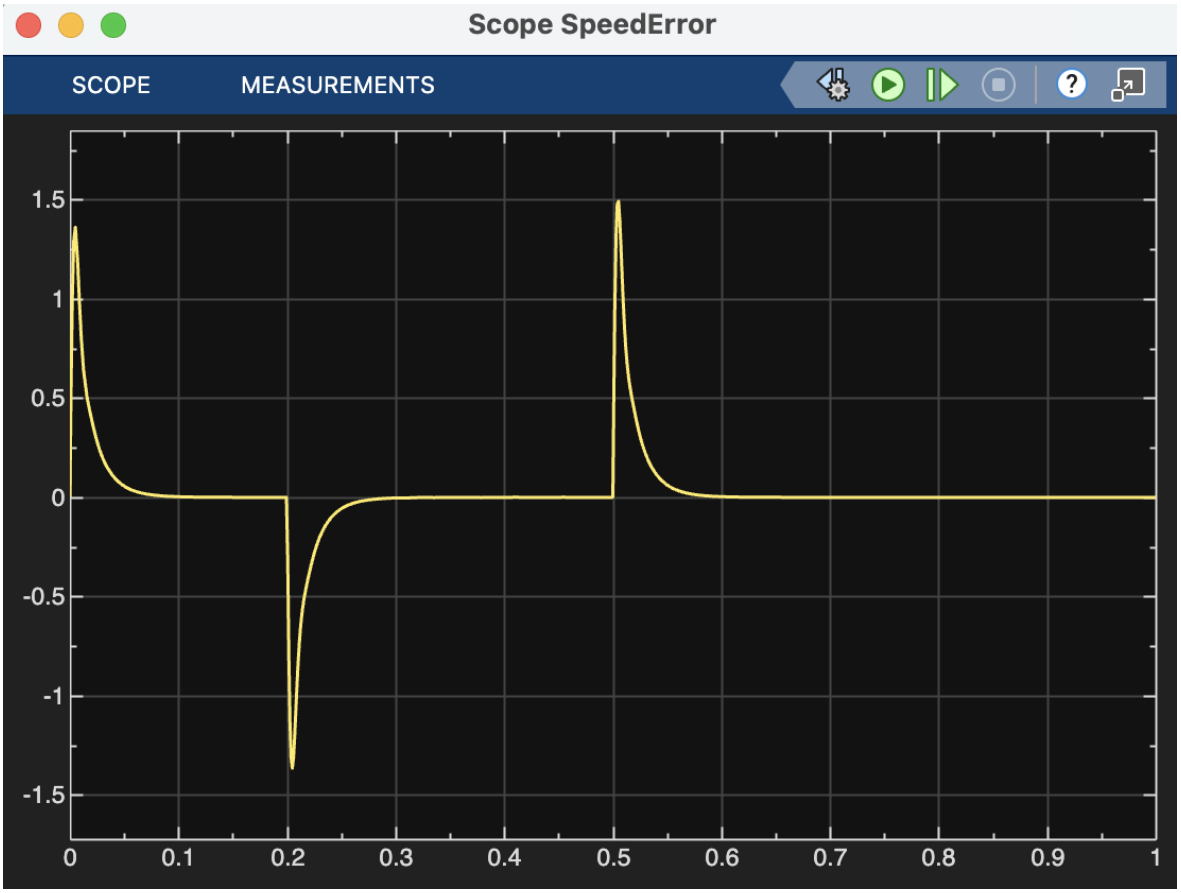


Figure 3. Scope of the speed tracking error.

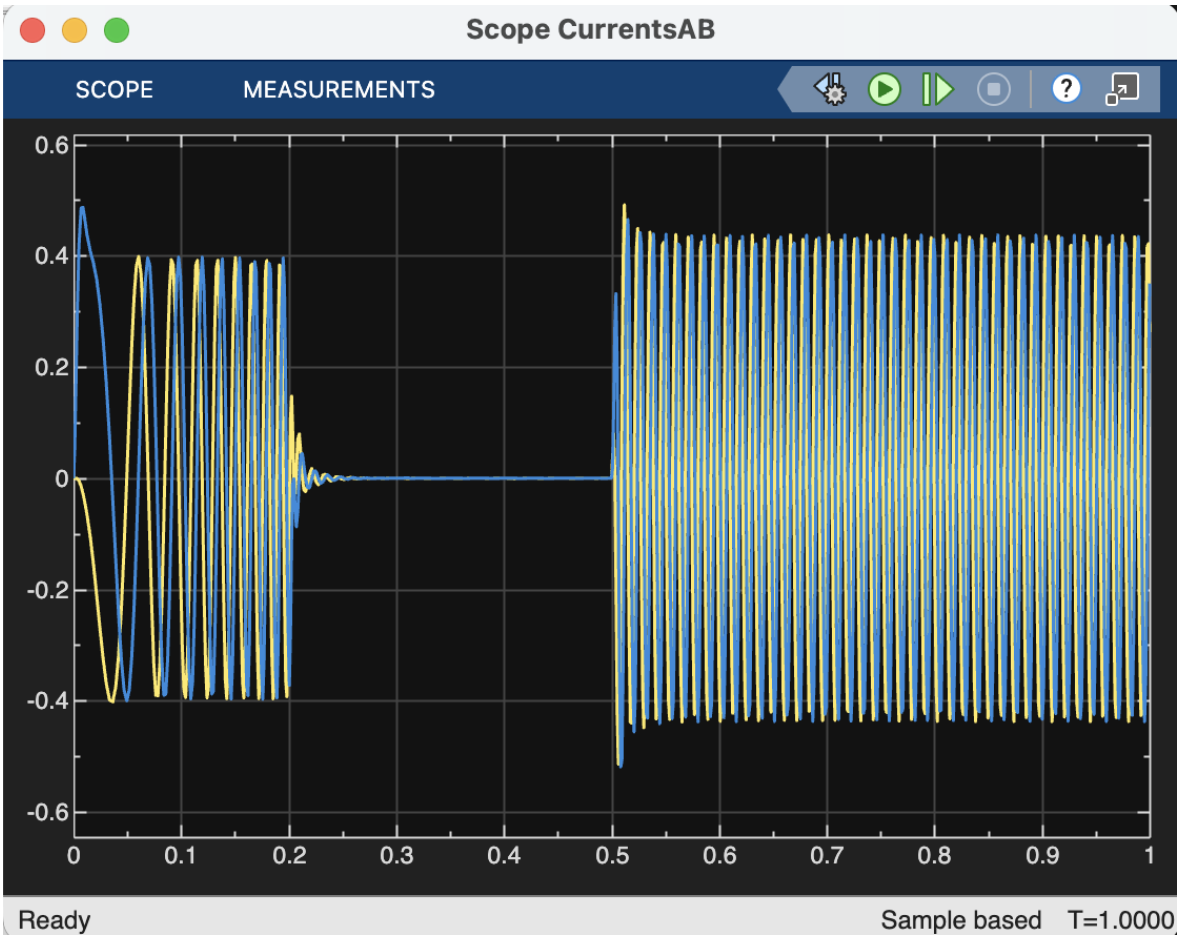


Figure 4. Scope of the stator currents in the alpha-beta frame.

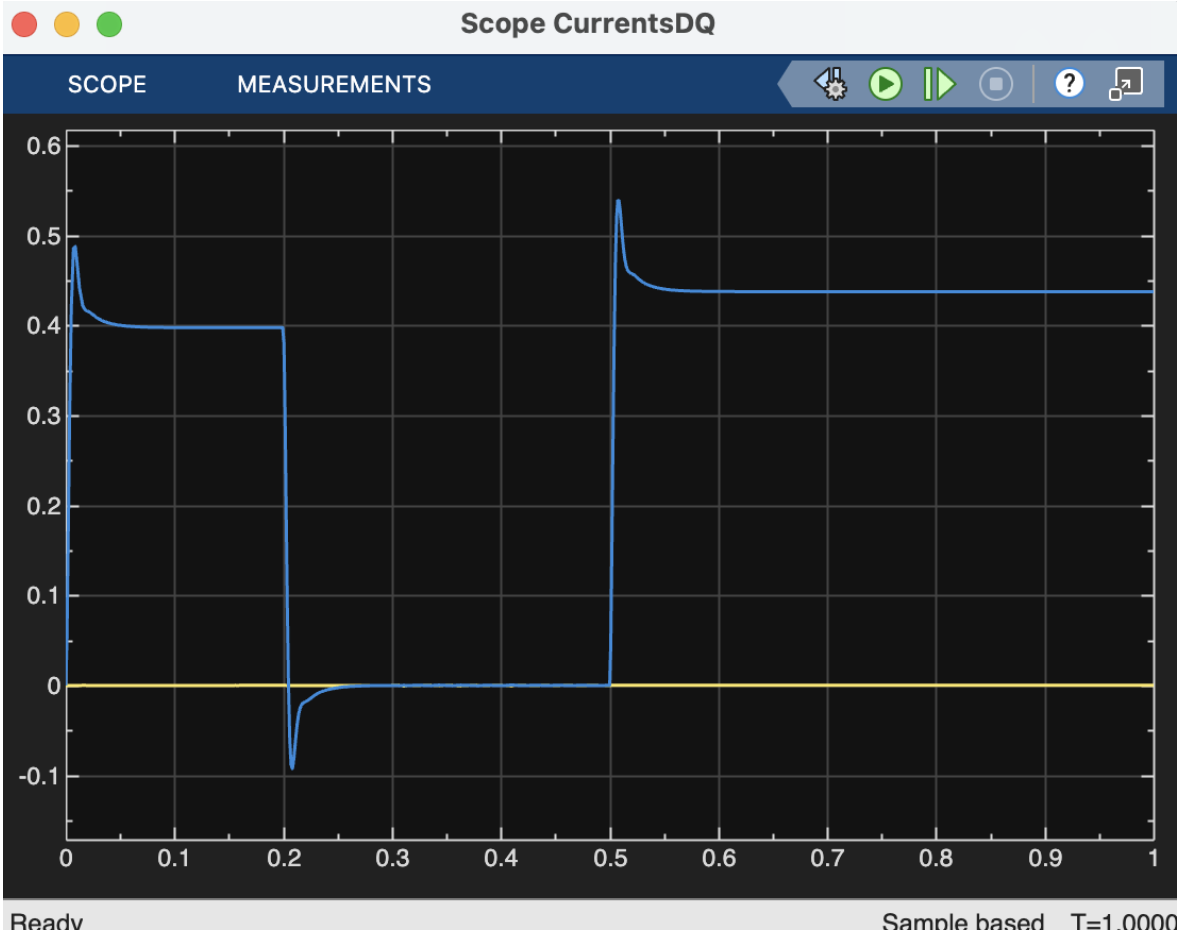


Figure 5. Scope of the stator currents in the dq frame.

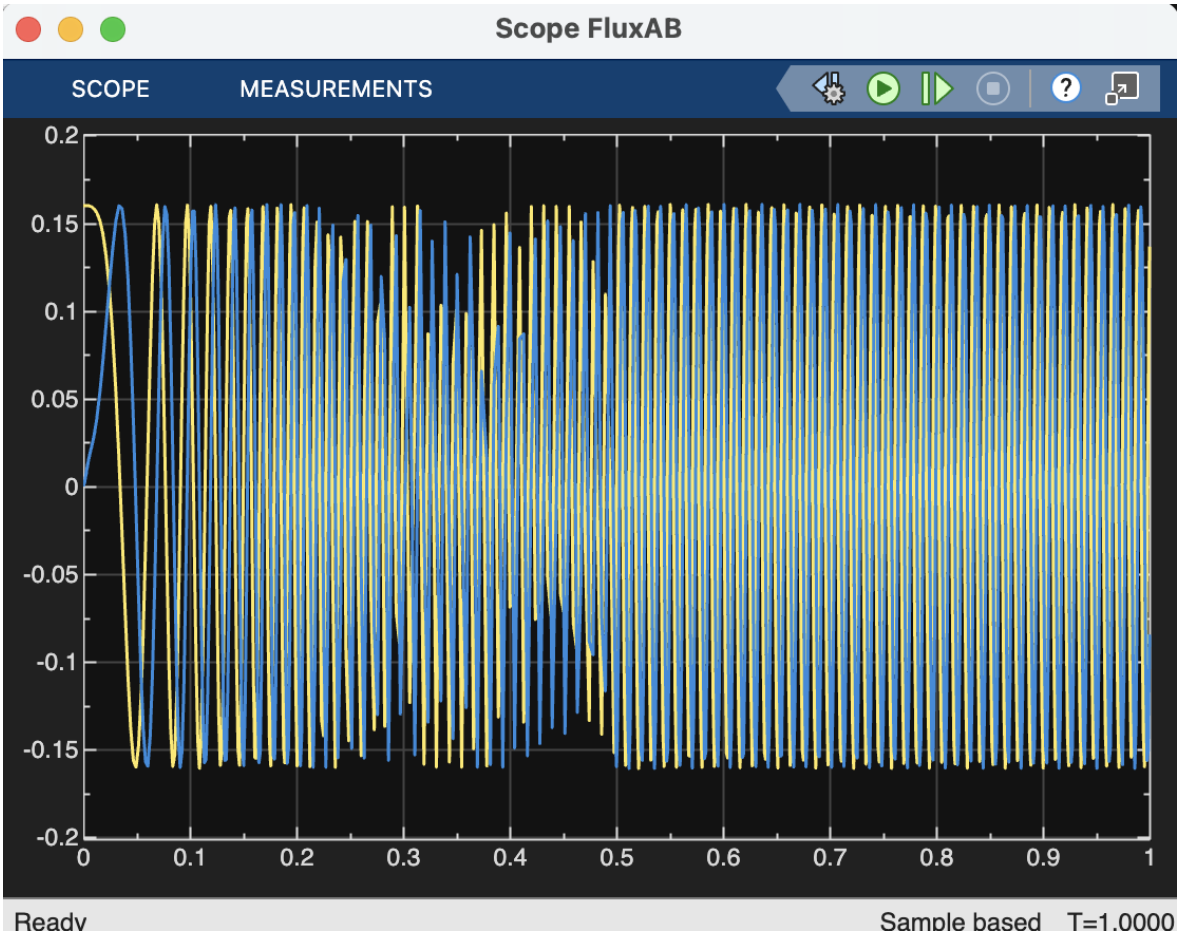


Figure 6. Scope of the total magnetic flux components in the alpha-beta frame.

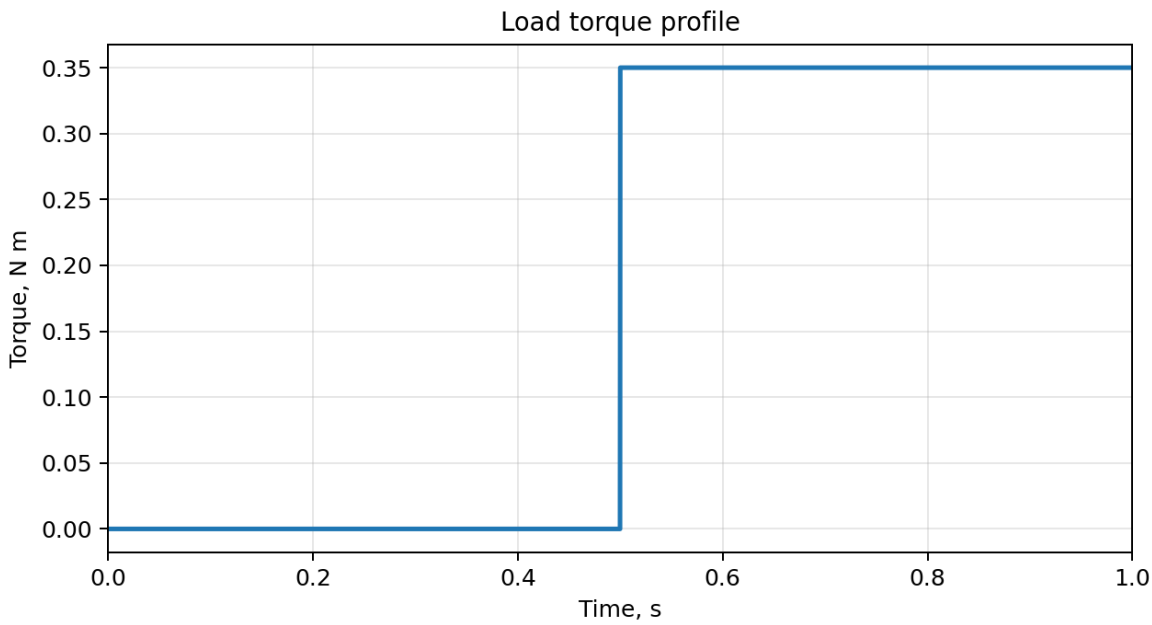


Figure 7. Load-torque profile used in the simulation.

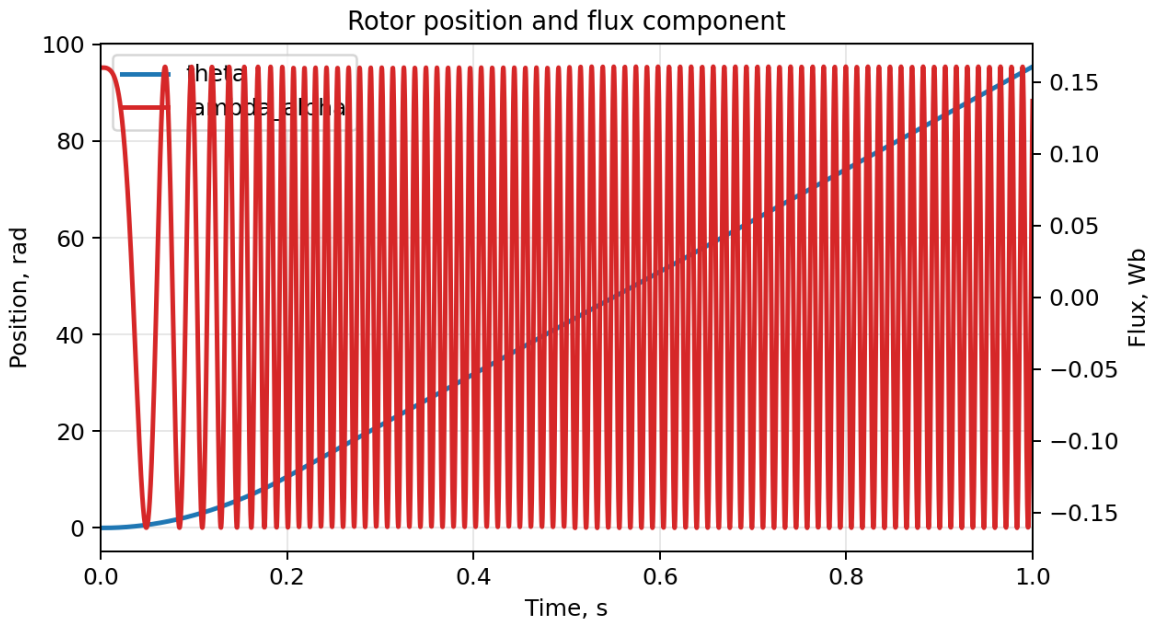


Figure 8. Rotor position and one flux component obtained in the numerical model.

# 4. Performance Summary

| Quantity | Value |
|---|---|
| omega*(0.4), rad/s | 106.000000 |
| omega*(1), rad/s | 106.000000 |
| omega(0.4), rad/s | 106.000015 |
| omega(1), rad/s | 106.000000 |
| e_omega(0.4), rad/s | -0.000015 |
| e_omega(1), rad/s | 0.000000 |
| Overshoot, % | 0.003010 |

# 5. Analysis and Conclusion

The tuned FOC achieved an overshoot of 0.00% and a steady-state speed error below 0.000 rad/s, which satisfies the assignment limits for variant 3. The d-axis current remains near zero, which confirms proper field orientation, while the q-axis current provides the required torque.